



**Guangdong Technion**

Israel Institute of Technology

广东以色列理工学院

# **035033-Intro to Sensor-Integrated Systems**

## **Lecture IV: Signals and Noise in Measurement Systems**

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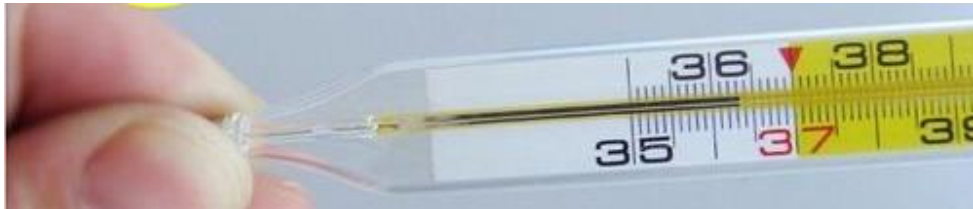
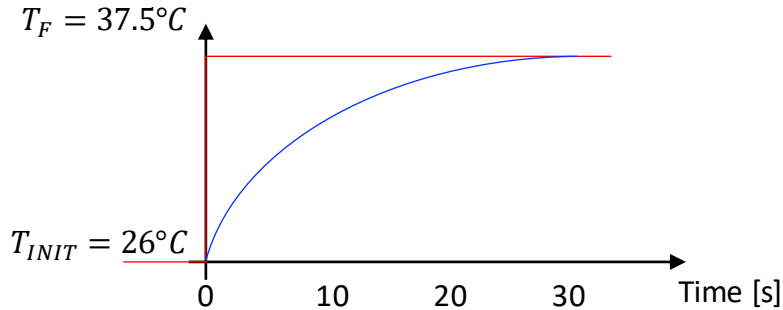
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# Dynamic Characteristics of Measurement Systems

- The ways in which an element responds to sudden input changes are termed its **dynamic characteristics**



# Intuitive understanding of time constant

- ❑ Time constant: 2 hours vs 63 hours
- ❑ Time constant larger  $\rightarrow$  slower
- ❑ For sensors, the smaller, the better

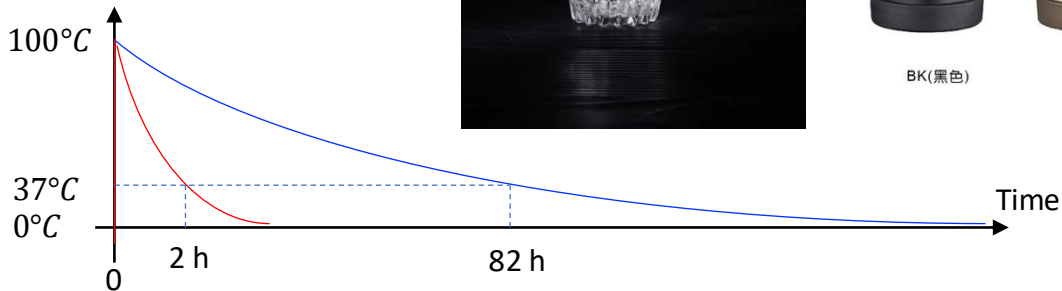


**THERMOS.** 膳魔師不銹鋼真空保溫杯  
TCMF-501 500ml

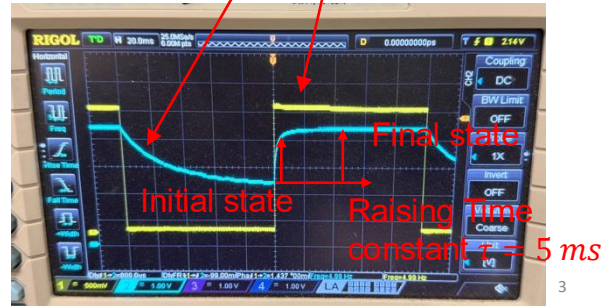
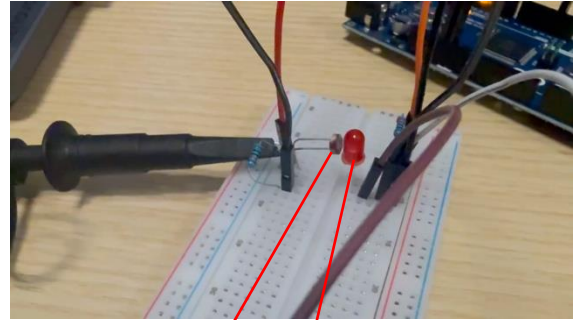
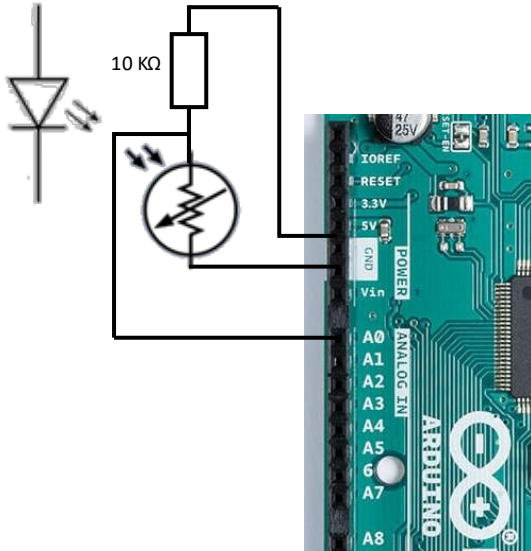


BK(黑色)

GL(金色)

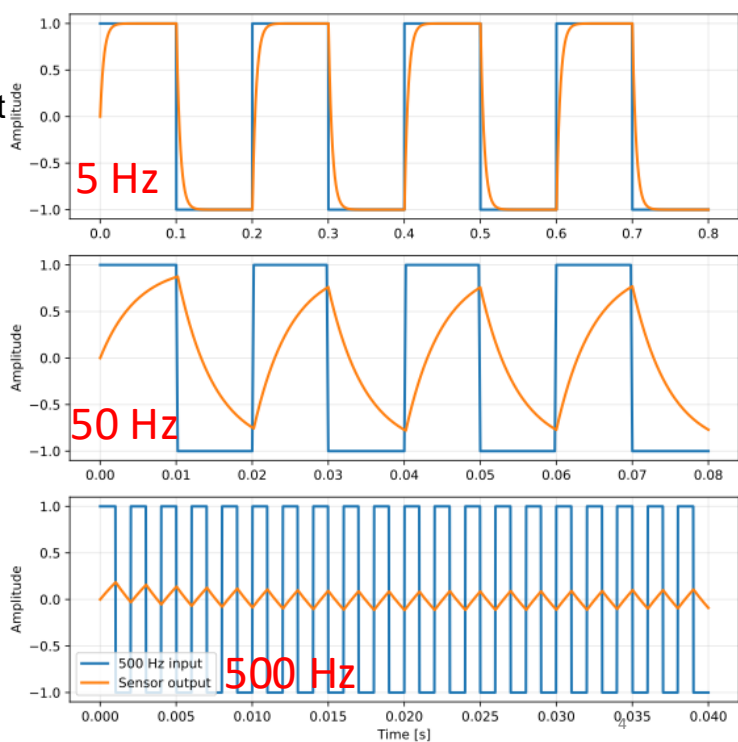


# Bandwidth: explain intuitively



## Take-home message

- ❑ No sensors are perfect;
- ❑ Each sensor have their own time constant
- ❑ Rule of thumb: good enough is enough



# Our eyes have their own time constant



## 2. Temporal resolution (perception level)

Instead of  $\tau$ , vision is often characterized by **critical flicker fusion (CFF)**:

- Typical human CFF:

$$f_{\text{CFF}} \approx 30\text{--}60 \text{ Hz}$$

This corresponds roughly to a time constant:

$$\tau \approx \frac{1}{2\pi f} \approx 3\text{--}10 \text{ ms}$$

👉 This is why:

- Movies at **24 fps** look continuous
- Screens use **60–120 Hz**

1 Hz



2 Hz



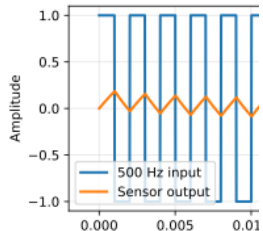
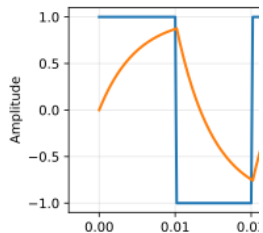
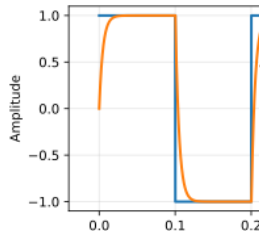
5 Hz



10 Hz

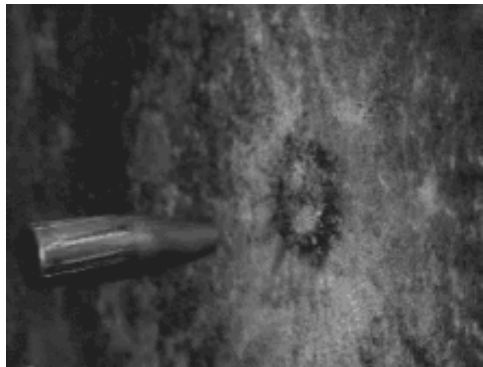


20 Hz

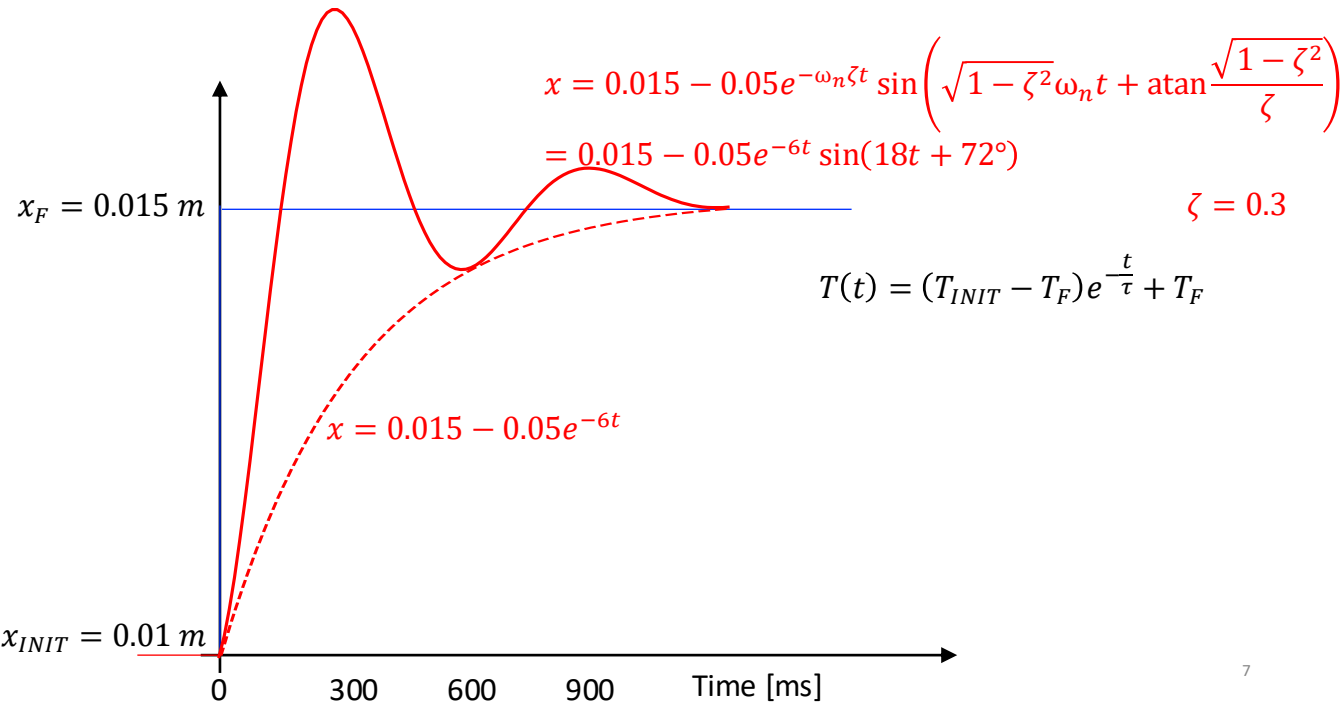


# Camera shuttle speed

- ❑ 🧑 Everyday / general photography: 1/60 – 1/200 s
- ❑ 🏃 Motion / robotics / human movement: 1/200 – 1/1000 s
- ❑ ⚡ Fast dynamics (sports, drones, impacts): 1/1000 – 1/5000 s (or faster)
- ❑ In response time, camera is much better than our eyes (10 ms vs. 0.2 ms)



For steady state,  $x_{INIT} = 0.01 \text{ m}$ ,  $x_F = 0.015 \text{ m}$





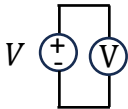
# Example of second order system

❑ Question: can you do 100 measurements within 1 second?



# Loading effect

- ❑ Ideal circuit  $V_M = V$



- ❑ Real circuit: Multimeter has a resistor in parallel

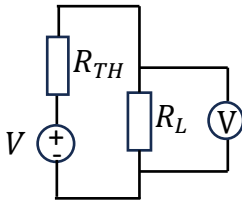
- ❑ Ideal model: assuming  $R_{TH} = 0 \Omega$ , and  $R_L = \infty \Omega$

- ❑ Real model:  $R_{TH} = 1 \Omega$ , and  $R_L = 1 M \Omega$

- ❑  $V_M = \frac{R_L}{R_{TH} + R_L} V = 0.9999999 V$

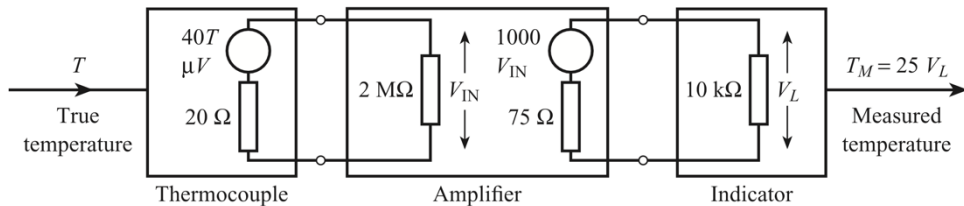
- ❑ This is not a problem when precision is not highly required

- ❑ However, to claim 0.00001% error, this is an issue



# Loading effect

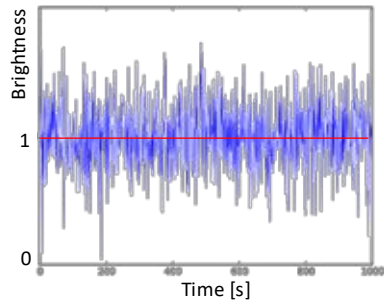
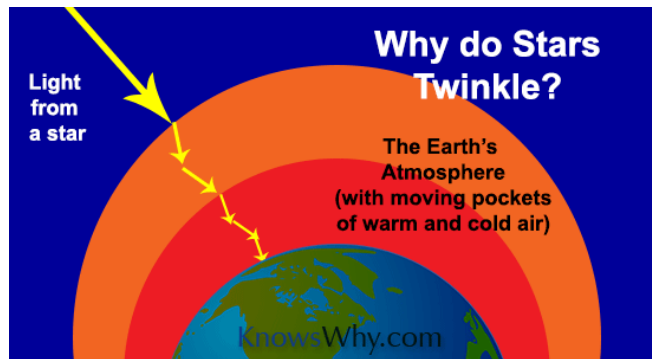
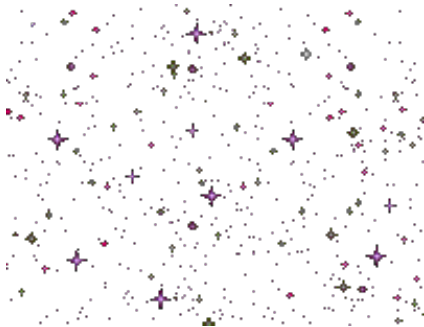
- ❑ Assumption: The external resistance  $R_L \gg R_{TH}$ , which is the internal resistance
- ❑  $R_L = 2\text{ M}\Omega \gg R_{TH} = 20\Omega$
- ❑ However, there is still a voltage divide effect: supposed to get  $40T$ , but in fact gets  $\frac{2000000}{2000000+20} 40T = \frac{2000000}{2000000+20} 40T = 0.99999 * 40T$
- ❑ There is  $1 - 0.99999 = 0.00001 = 0.001\%$  error.
- ❑ This should be considered if the precision requirement is equal or higher than  $0.001\%$



**Question:** what is the error in percentage of this instrument?

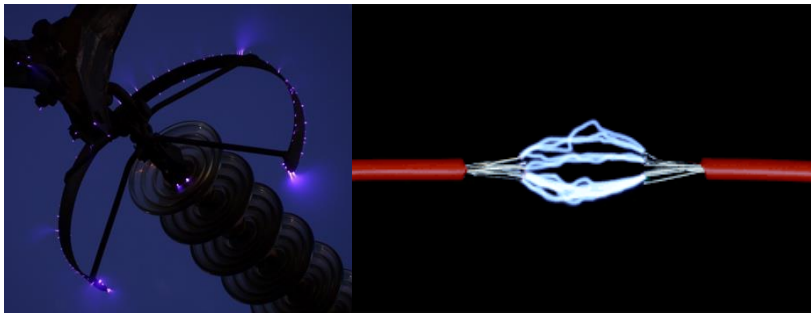
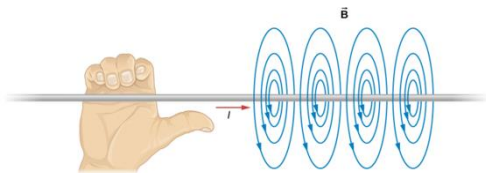
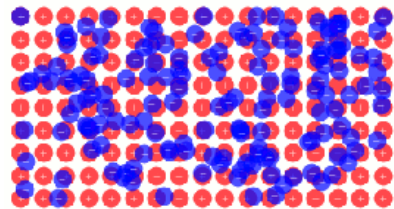
# Noise in measurement

- ❑ Noise is unavoidable in measurement



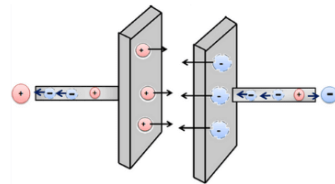
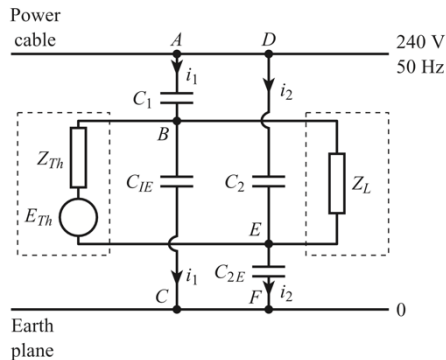
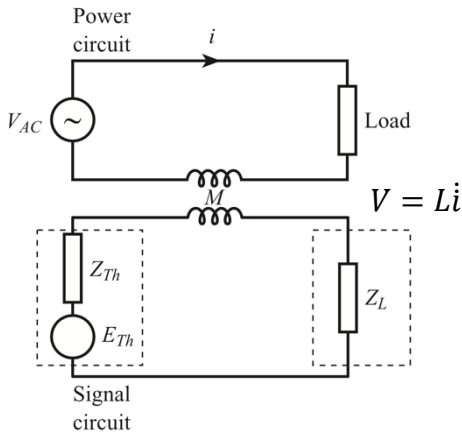
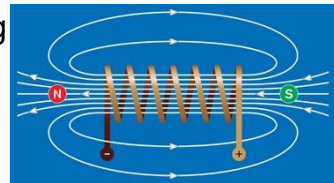
# Sources of noise

- ❑ Random motion of electrons
- ❑ External interference: nearby A.C. power circuits which usually operate at 240 V, 50 Hz.
- ❑ Turning on and off: large current change
- ❑ Corona discharge (电晕放电) in the vicinity of high voltage power circuits
- ❑ Fluorescent lighting (arcing 电弧)
- ❑ Radio-frequency transmitters  
welding equipment  
and electric arc furnaces



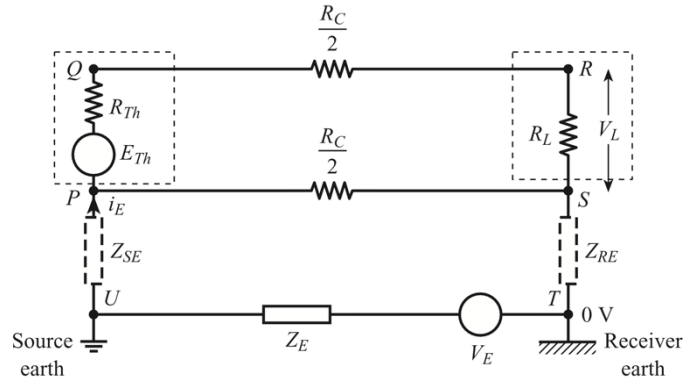
# Coupling mechanisms to external sources

- ❑ **Inductive or Electromagnetic coupling:** by surrounding magnetic field
- ❑ **Capacitive or electrostatic coupling**



# Multiple earth

- ❑ Ideally  $Z_{SE} = 0$ ,  $Z_{RE} = 0$ ,  $Z_E = 0$ ,  $V_E = 0$ .
- ❑ However, in reality, those are not zero and becomes interference

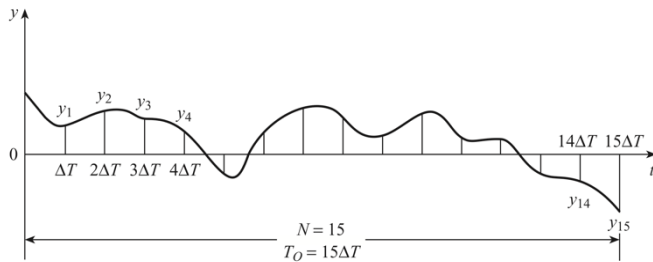


# Random signal: description

- Sampling period  $\Delta T = \frac{1}{f}$ , where  $f$  is the sampling frequency

## Description

- Mean  $\bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$
- Standard deviation  $\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \bar{y})^2}$
- Root mean square  $y_{rms} = \sqrt{\frac{1}{N} \sum_{i=1}^N y_i^2}$





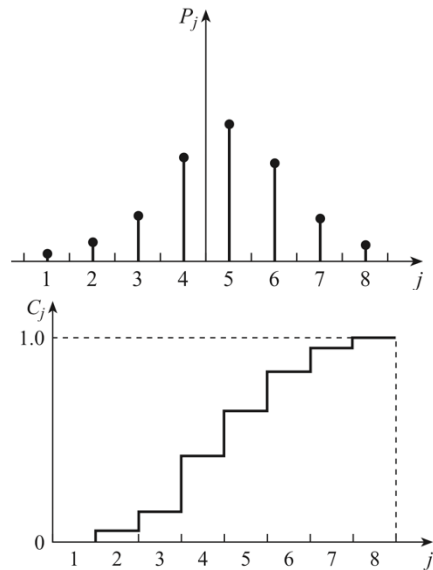
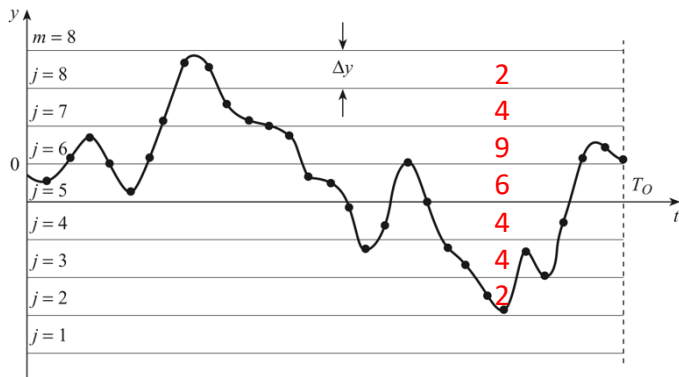
# Probability density function

□ The probability of the signal occurring in the  $j^{\text{th}}$  section:  $P_j = \frac{\text{number of samples in } j^{\text{th}} \text{ section}}{\text{total number of samples}} =$

$$\frac{n_j}{N}$$

□ Cumulative probability  $C_j = P_1 + P_2 + \dots + P_j$

□  $C_N = 1$

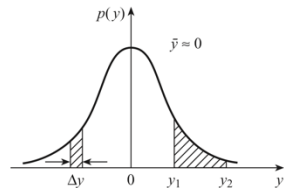
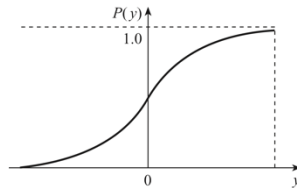
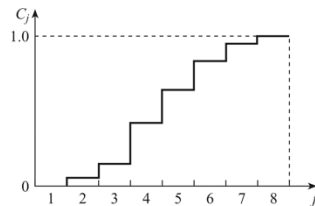
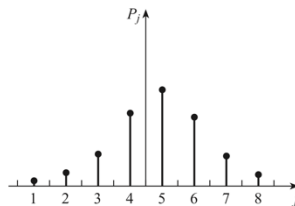
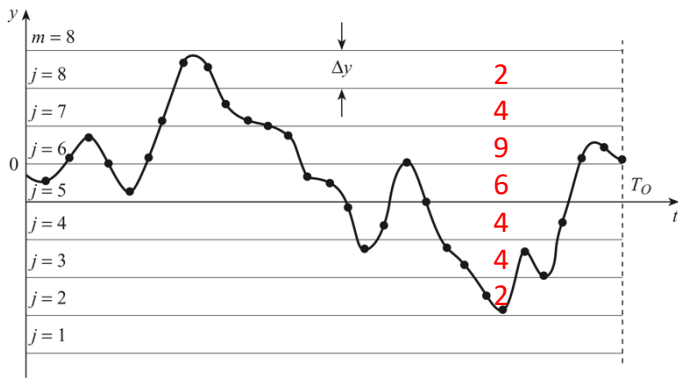


# Probability density function

□ When  $\Delta y \rightarrow 0$ :

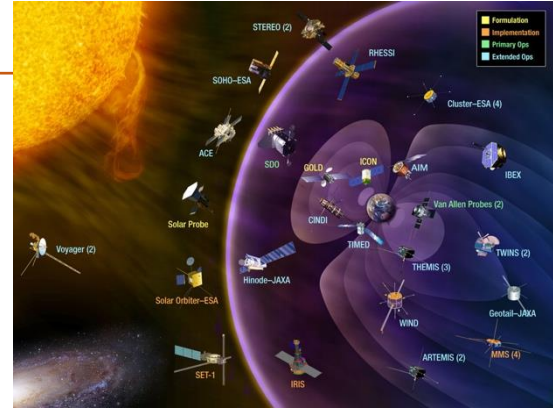
□  $P_j \Rightarrow p(y)$

□  $C_j \Rightarrow P(y)$



# Avoid noise in measurement

- ☐ Physical method
- ☐ Do as much as possible

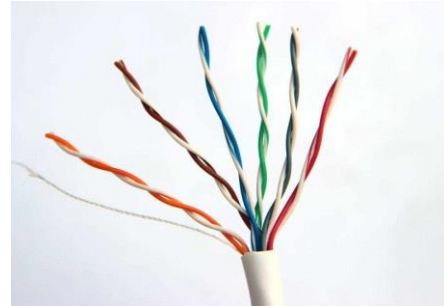
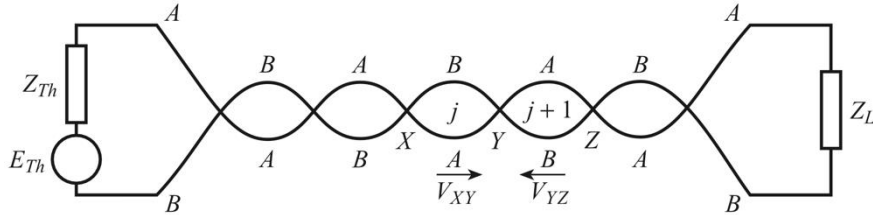
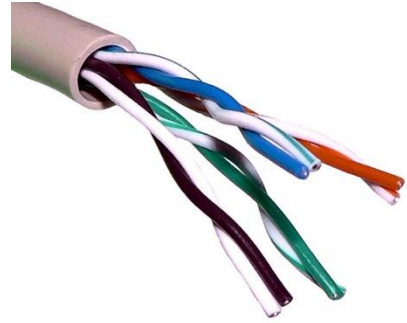


Plateau of high altitude (thin air)  
No ambient light

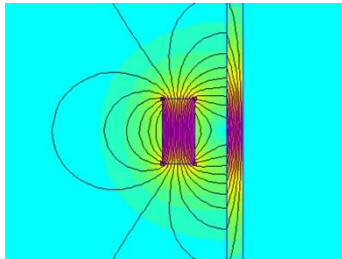
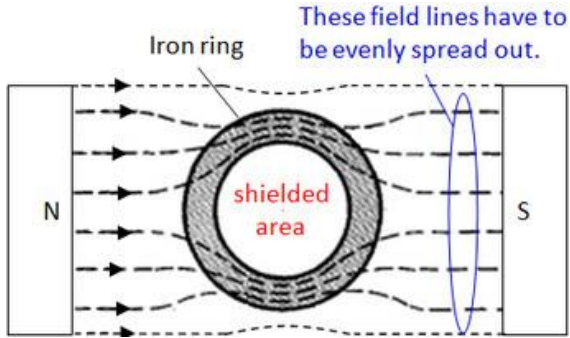


# Methods of reducing effects of noise and interference

- ❑ Electromagnetic shielding
- ❑ Twisted pairs of cable



# Shielding effect and shielding web



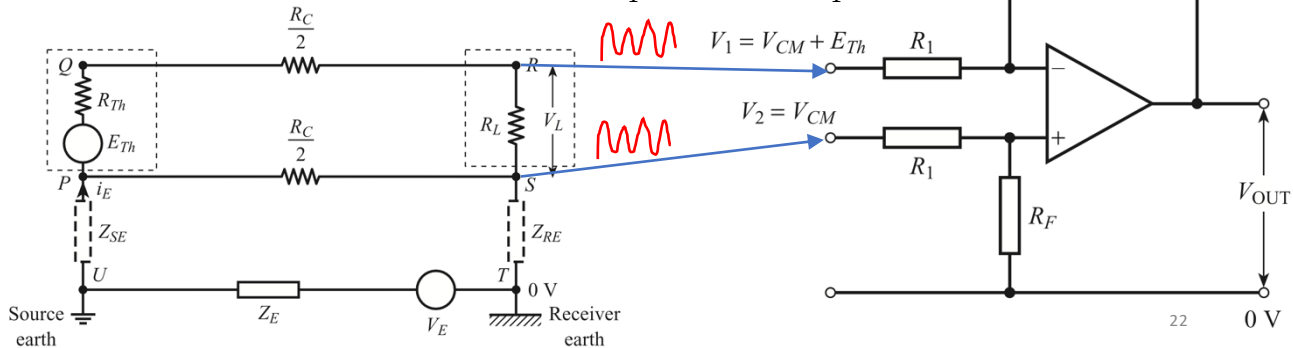
## Differential amplifier kill common mode interference

$$\boxed{V_+ = \frac{R_F}{R_1 + R_F} V_2}$$

Virtual short:  $V_- = V_+ = \frac{R_F}{R_1 + R_F} V_2$

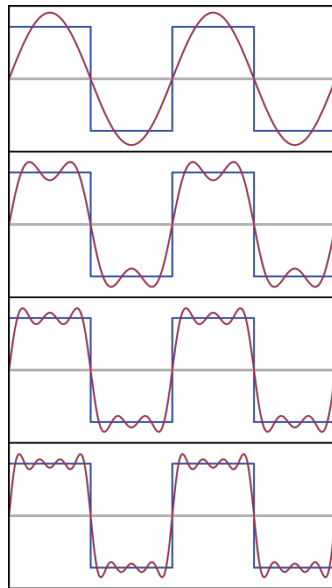
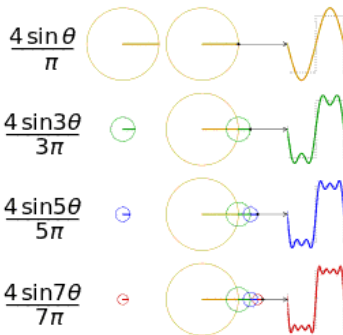
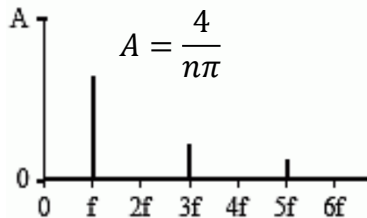
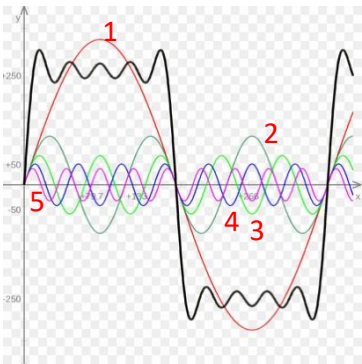
❑ Virtual open:  $I_{+-} = 0$

$\square$  KCL  $\frac{V_1 - V_-}{R_1} = \frac{V_- - V_{OUT}}{R_F} \Rightarrow \frac{V_1 - \frac{R_F}{R_1 + R_F} V_2}{R_1} = \frac{\frac{R_F}{R_1 + R_F} V_2 - V_{OUT}}{R_F} \Rightarrow R_1 R_F V_1 + R_F R_F V_1 - R_F R_F V_2 = R_1 R_F V_2 - R_1 R_1 V_{OUT} - R_F R_1 V_{OUT} \Rightarrow V_{OUT} = \frac{R_F}{R_1} (V_2 - V_1) = \frac{R_F}{R_1} E_{Th}$



# Fourier series

Unit square wave  $s(t) = \sum_{n=1,3,5,\dots}^N \left( \frac{4}{n\pi} \sin(n\omega t) \right)$



$$\frac{4}{\pi} \sin(\omega t)$$

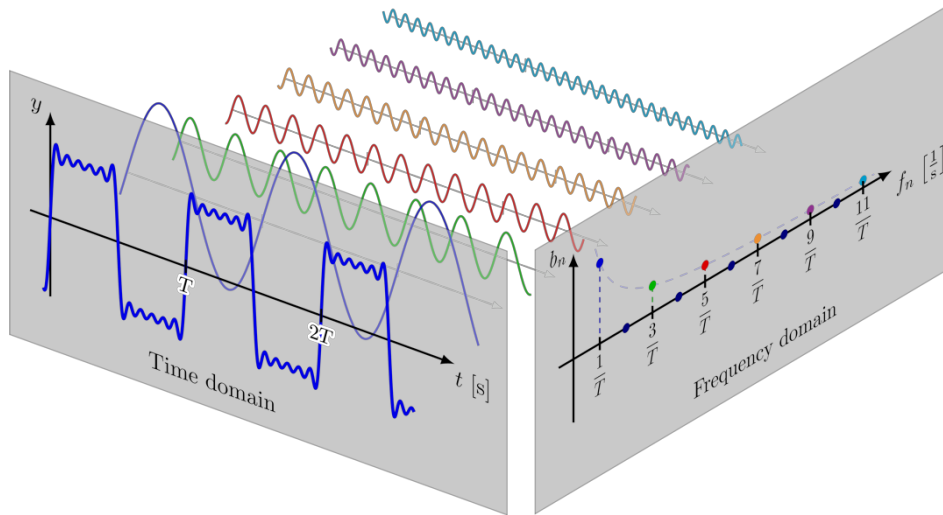
$$\frac{4}{\pi} \sin(\omega t) + \frac{4}{3\pi} \sin(3\omega t)$$

$$\frac{4}{\pi} \sin(\omega t) + \frac{4}{3\pi} \sin(3\omega t) + \frac{4}{5\pi} \sin(5\omega t)$$

$$\frac{4}{\pi} \sin(\omega t) + \frac{4}{3\pi} \sin(3\omega t) + \frac{4}{5\pi} \sin(5\omega t) + \frac{4}{7\pi} \sin(7\omega t)$$

# Fourier series of square wave

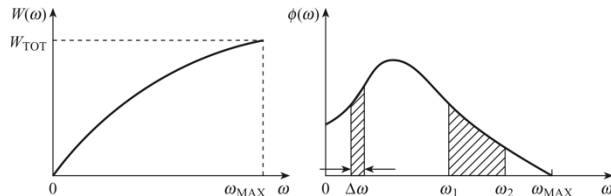
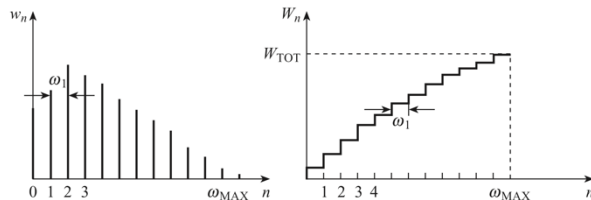
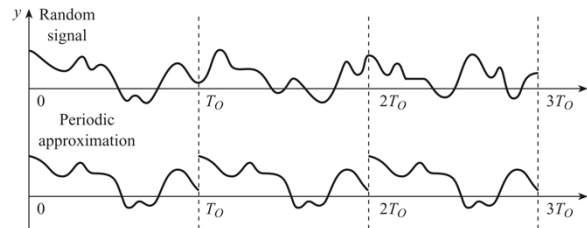
$$\square F(t) = \sum_{n=1,3,5,\dots}^{\infty} \frac{4}{n\pi} \sin n\omega t$$





# Power spectral density $\phi(\omega)$

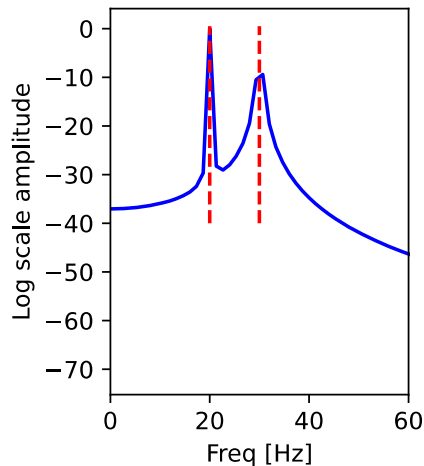
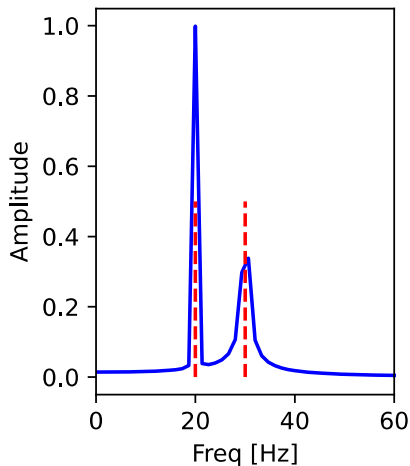
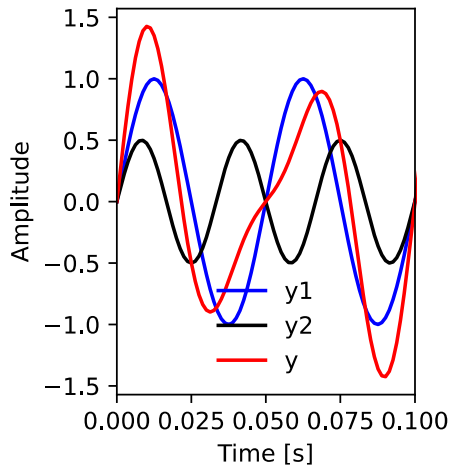
- ❑ Fourier series of random signal (signal)  $y(t) = a_0 + \sum_{i=1}^N \left( a_n \cos \frac{n}{N} \omega_1 t + b_n \sin \frac{n}{N} \omega_1 t \right)$
- ❑ If the resistance is  $1 \Omega$ , then power is  $w_n = \frac{1}{2} (a_n^2 + b_n^2)$
- ❑  $W_n = w_1 + w_2 + w_3 + \dots + w_n$
- ❑  $W(\omega) = \lim_{n \rightarrow \infty} W_n$
- ❑ Power spectral density (PSD)  $\phi(\omega) = \frac{dW(\omega)}{d\omega}$



# Fourier Transfer Example

$$y_1 = \sin(20 * 2\pi t)$$

$$y_2 = 0.5 * \sin(30 * 2\pi t),$$



# Code for you to play

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.fft import fft, fftfreq
```

```
N = 600 # Number of sample points
T = 1 / 800 # sample spacing
t = np.linspace(0.0, N*T, N)
y1 = np.sin(20 * 2.0*np.pi*t)
y2 = 0.5*np.sin(30 * 2.0*np.pi*t)
y = y1+y2
yf = fft(y)
xf = fftfreq(N, T)[:N//2]
```

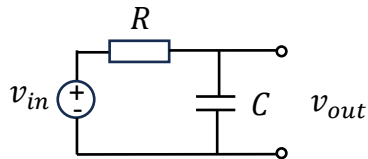
```
plt.figure(figsize=[8, 3])
plt.subplot(131)
plt.plot(t, y1, 'b', label='y1')
plt.plot(t, y2, 'k', label='y2')
plt.plot(t, y, 'r', label='y')
plt.xlim([0, 0.1])
plt.legend()
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
plt.subplot(132)
plt.plot(xf, 2.0/N * np.abs(yf[0:N//2]), 'b')
plt.plot([20, 20], [0, 0.5], 'r--')
plt.plot([30, 30], [0, 0.5], 'r--')
plt.xlim([0, 60])
plt.xlabel('Freq [Hz]')
plt.ylabel('Amplitude')
plt.subplot(133)
plt.plot(xf, 20*np.log10(2.0/N * np.abs(yf[0:N//2])), 'b')
plt.plot([20, 20], [-40, 0.5], 'r--')
plt.plot([30, 30], [-40, 0.5], 'r--')
plt.xlim([0, 60])
plt.xlabel('Freq [Hz]')
plt.ylabel('Log scale amplitude')
plt.tight_layout()
plt.show()
```

# First order filtering

❑ System equation:  $R_T C \dot{v}(t) + v(t) = v_{in} = Y \cos(\omega t)$

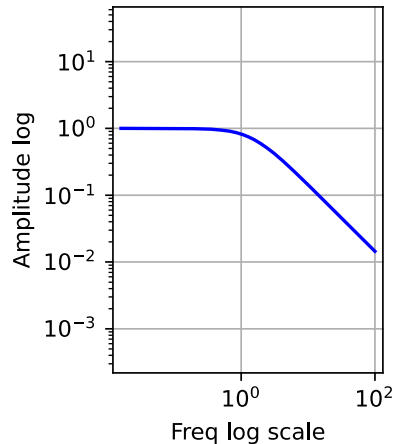
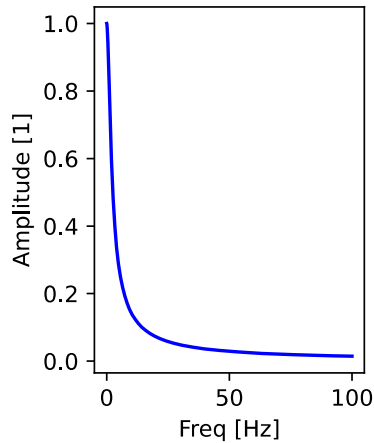
❑ Solution:  $v(t) = \frac{v_{in}}{1 + R_T C j\omega} = \frac{Y}{\sqrt{1 + R_T^2 C^2 \omega^2}} \cos\left(\omega t - \arcsin\left(\frac{R_T C \omega}{\sqrt{1 + R_T^2 C^2 \omega^2}}\right)\right)$

❑ Gain =  $\frac{1}{\sqrt{1 + R_T^2 C^2 \omega^2}}$



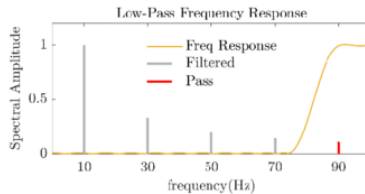
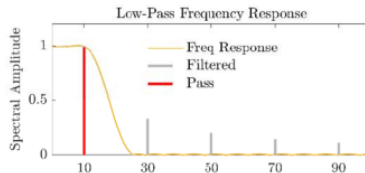
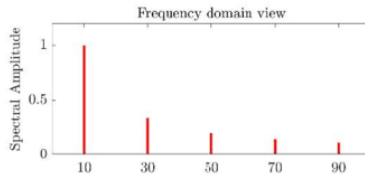
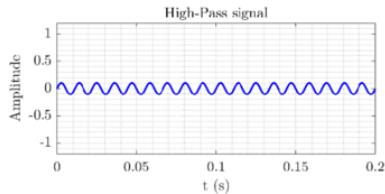
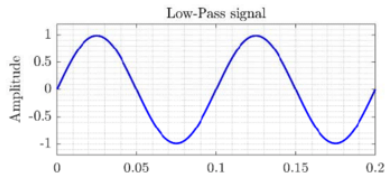
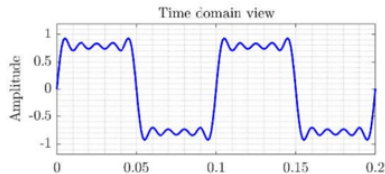
# Frequency response

$$\square \text{ Gain} = \frac{1}{\sqrt{1+R_T^2 C^2 \omega^2}}$$



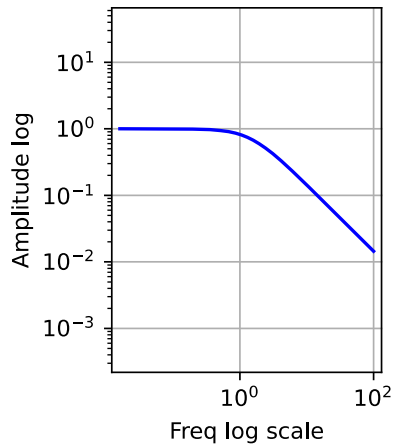
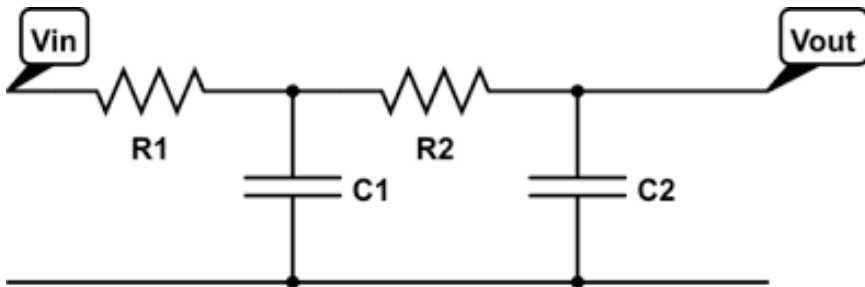
# How does filtering happen

## □ How pass filter vs. high pass filter

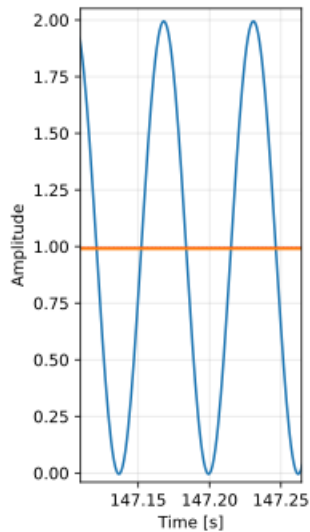
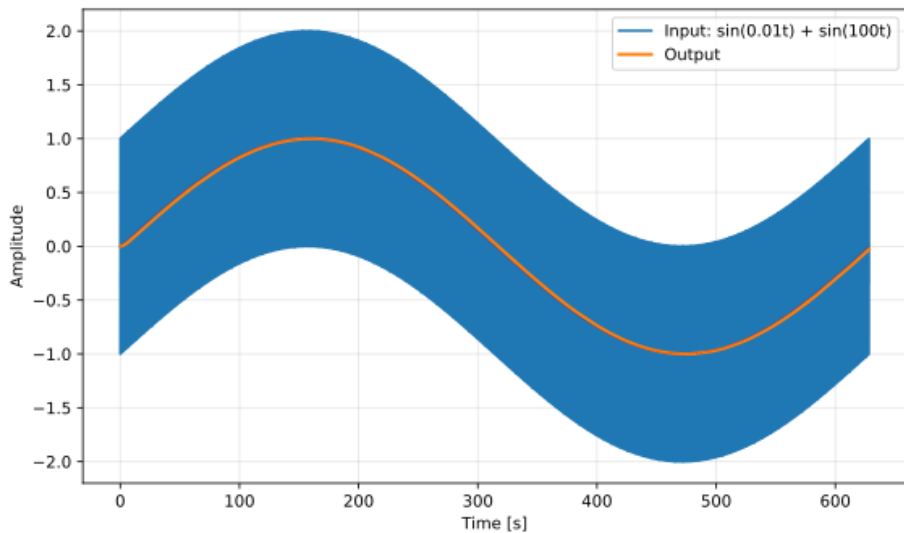


# Exercise

- Find the Gain of the following circuit
- $R_1 = R_2 = 1 \Omega$ ,  $C_1 = C_2 = 1 H$ .
- Amplitude of  $V_{out}$  when  $V_{in}$  has frequency of  $0.01 \text{ rad/s}$
- Amplitude of  $V_{out}$  when  $V_{in}$  has frequency of  $100 \text{ rad/s}$



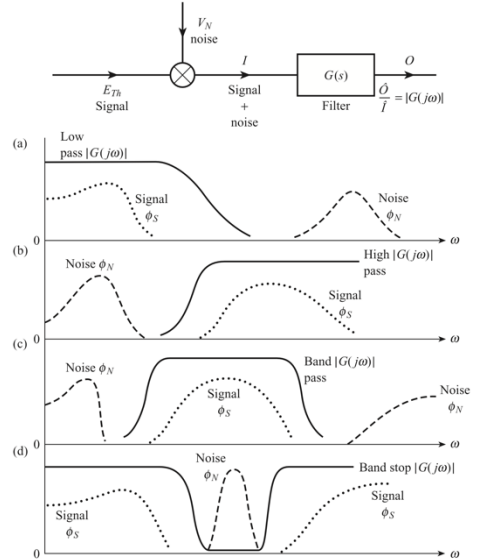
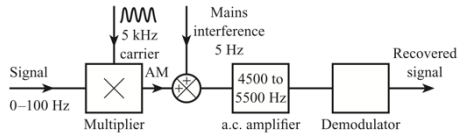
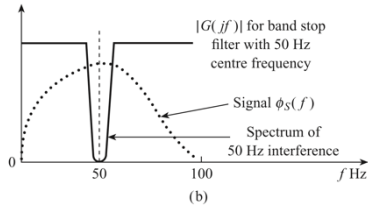
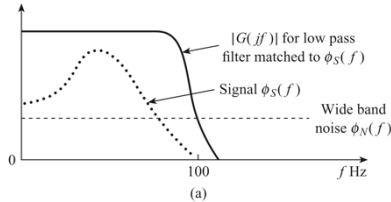
# Illustration





# Examples

- ❑ Low pass filter
- ❑ Band pass filter



# Filter example

$$\square \text{ Transfer Function} = \frac{300000000}{s^5 + 100s^4 + 10000s^3 + 600000s^2 + 20000000s + 300000000}$$

# Load sample data from a WAV file

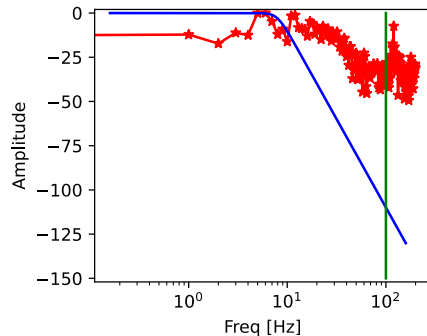
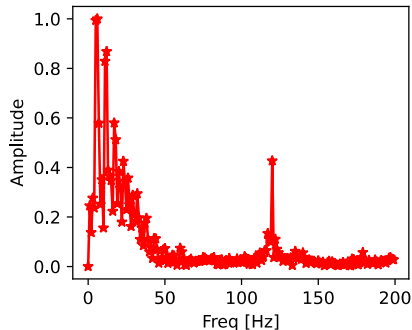
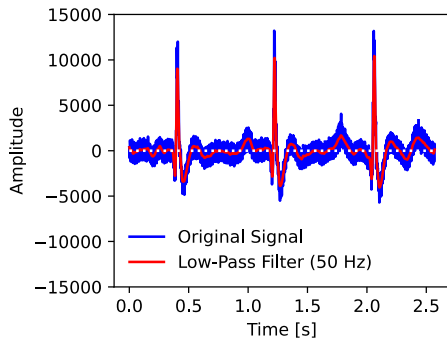
```
sample_rate, data = scipy.io.wavfile.read('ecg.wav')
```

```
times = np.arange(len(data))/sample_rate
```

# Apply a 50Hz low-pass filter to the original data

```
cutoff = 50; poles = 5
```

```
sos = scipy.signal.butter(poles, cutoff, 'lowpass', fs=sample_rate, output='sos')
```



```

import numpy as np
import scipy.io.wavfile
import matplotlib.pyplot as plt

# Load sample data from a WAV file
[sample_rate, temp] = scipy.io.wavfile.read('ecg.wav')
data = temp - np.mean(temp)
t = np.arange(len(data))/sample_rate

# Apply a 50Hz low-pass filter to the original data
cutoff = 50
poles = 5
Filter = scipy.signal.butter(poles, cutoff, 'lowpass', fs=sample_rate,
output='sos')
[b, a] = scipy.signal.butter(poles, cutoff, 'low', analog=True)
[w, h] = scipy.signal.freqs(b, a)
filtered = scipy.signal.sosfiltfilt(Filter, data)

# get frequency response
N = len(data)
M = 200
freqDetect = np.zeros((M, N), dtype=complex)
for i in range(M):
    freqDetect[i, 0:N] = np.sin(2 * i * np.pi * t.T) + 1j * np.cos(2 * i * np.pi * t.T)
zeros_save = np.zeros(np.shape(t))
imageResult = np.abs(np.dot(freqDetect, data))

```

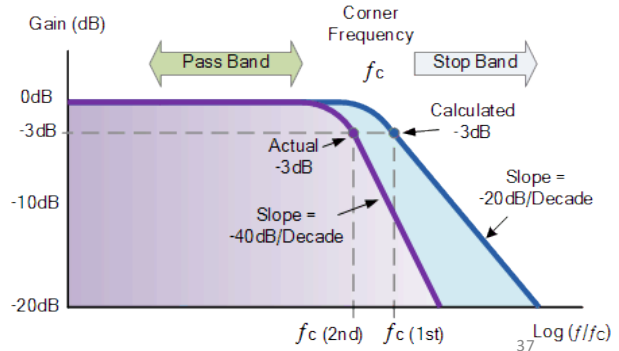
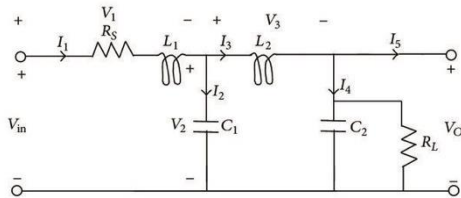
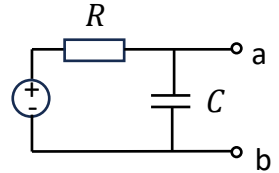
```

# Code used to display the result
plt.figure(figsize=(11, 3))
plt.subplot(131)
plt.plot(t, data, 'b', label='Original Signal')
plt.plot(t, filtered, 'r', label='Low-Pass Filter (50 Hz)')
plt.plot(t, np.zeros(np.shape(filtered)), 'w:')
plt.legend()
plt.xlabel('Time [s]')
plt.ylabel('Amplitude')
plt.ylim([-15000, 15000])
plt.subplot(132)
plt.plot(range(M), imageResult/np.max(imageResult), 'r-*')
plt.xlabel("Freq [Hz]")
plt.ylabel("Amplitude")
plt.subplot(133)
plt.plot(range(M), 20 *
np.log10(abs(imageResult)/np.max(imageResult)), 'r-*')
plt.semilogx(w/2/np.pi, 20 * np.log10(abs(h)), 'b')
plt.plot([100, 100], [-150, 0], color='green')
plt.ylim([-152, 2])
plt.xlabel("Freq [Hz]")
plt.ylabel("Amplitude")
plt.tight_layout()
plt.show()
plt.savefig('temp.pdf')

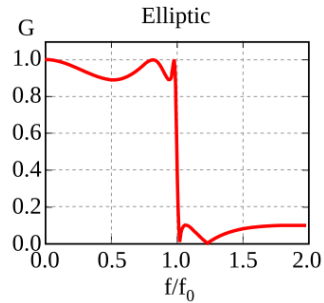
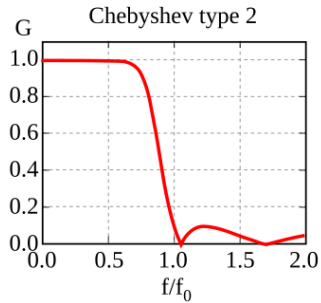
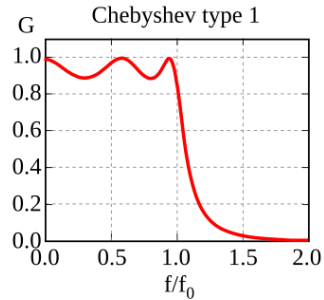
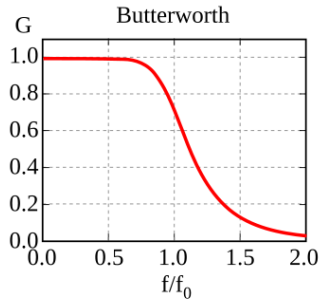
```

# High order filtering

- ❑ First order:  $Gain = \frac{1}{\sqrt{1+R_T^2 C^2 \omega^2}}$
- ❑ Second order:  $Gain = \frac{1}{\sqrt{1+\dots+\omega^4}}$
- ❑ Fourth order:  $Gain = \frac{1}{\sqrt{1+\dots+\omega^8}}$



# Different types of filters

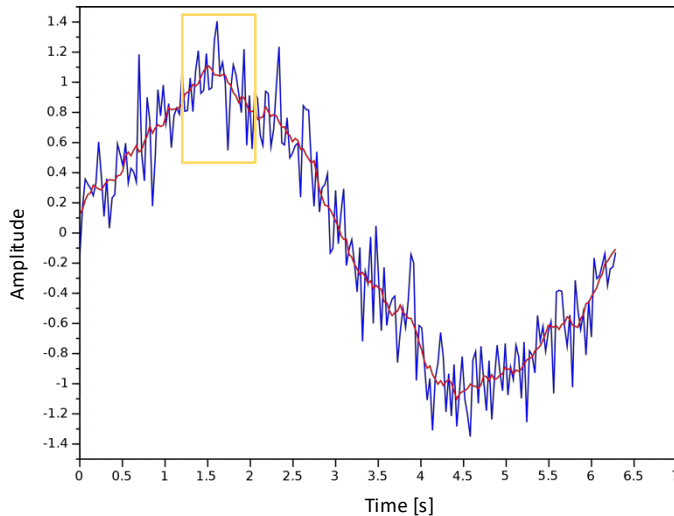


[https://en.wikipedia.org/wiki/Chebyshev\\_filter](https://en.wikipedia.org/wiki/Chebyshev_filter)

# Moving average

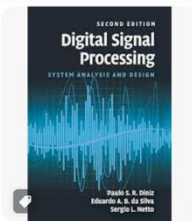
□ Simple moving average (SMA) is the unweighted mean of the previous  $k$  data-points

$$\begin{aligned} SMA_k &= \frac{p_{n-k+1} + p_{n-k+2} + \cdots + p_n}{k} \\ &= \frac{1}{k} \sum_{i=n-k+1}^n p_i \end{aligned}$$

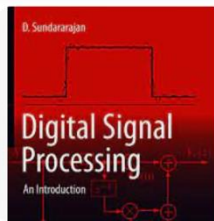


# Digital Singal Processing

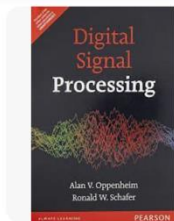
❑ A whole book and a full semester for this topic



Amazon.com · In stock  
Digital Signal Processi...



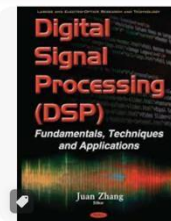
SpringerLink  
Digital Signal Processing: ...



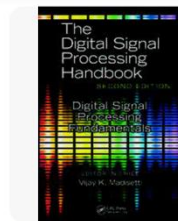
Amazon.in  
Digital Signal Processi...



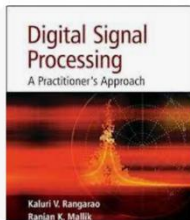
SpringerLink  
Principles of Digital Si...



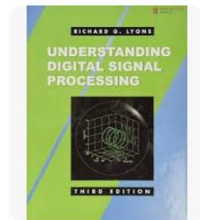
Nova Science... · In stock  
Digital Signal Processi...



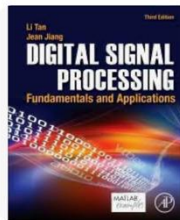
Routledge  
Digital Signal Processi...



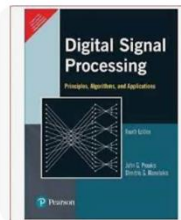
Wiley  
Digital Signal Processin...



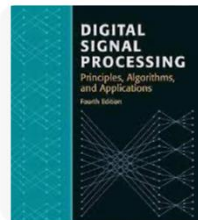
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IndiaMART  
Digital Signal Processi...



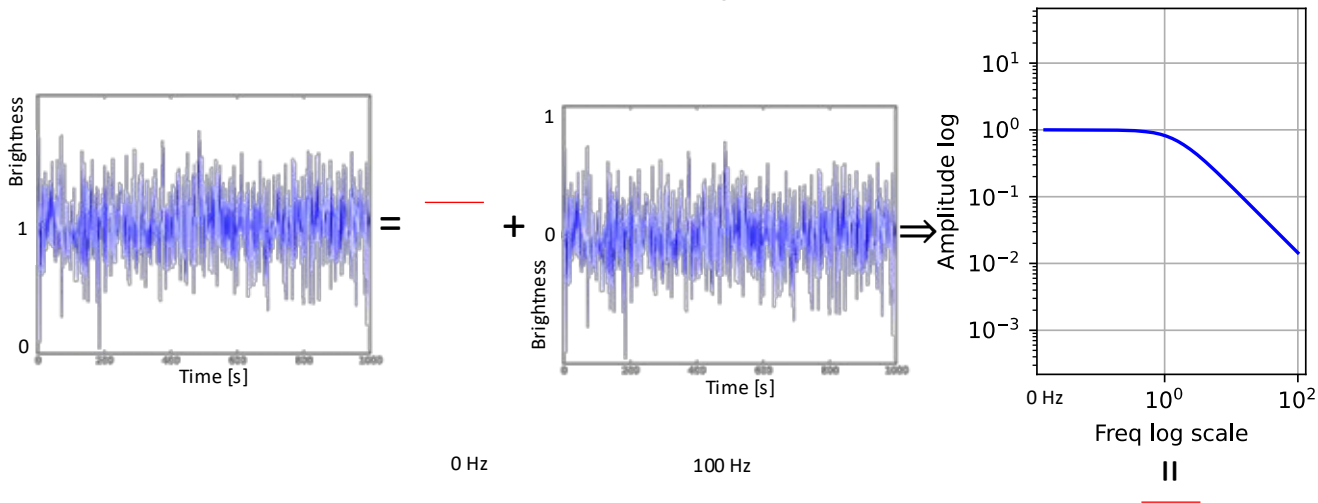
Goodreads  
Digital Signal Processing...



People  
DIGITAL SIGNAL PROC...

# Take home message

- ❑ Signal and noise have different characteristics – different frequencies
- ❑ Filter is used to kill the noise while retain the signal





---

Table Prob. 1 gives 50 sample values of a random signal.

- (a) Estimate the mean and standard deviation of the signal.
- (b) Using an interval  $\Delta y = 0.5$  V, draw the  $P_j$  and  $C_j$  discrete probability distributions for the signal.

---

|       |       |       |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| −0.59 | 1.02  | −0.25 | −0.34 | 0.95  | 1.24  | −0.30 | 0.21  | −0.89 |
| −1.00 | 1.36  | 0.03  | 0.04  | −0.13 | −0.71 | −1.23 | 0.03  | −1.00 |
| 0.65  | 0.11  | 0.99  | 0.17  | 0.39  | 2.61  | −0.08 | −0.33 | 0.99  |
| 2.15  | 0.91  | 0.89  | 1.43  | −1.69 | −0.25 | 2.47  | −1.97 | −2.26 |
| 0.42  | 0.05  | 0.26  | 0.33  | −0.42 | 0.79  | −0.07 | −0.32 | −0.66 |
| −0.63 | −0.06 | −0.61 | 0.77  | 1.90  |       |       |       |       |

---



Two complete periods of a square wave can be represented by the following 20 sample values:

+1+1+1+1+1      -1-1-1-1-1      +1+1+1+1+1      -1-1-1-1-1

Find the autocorrelation function of the signal over one complete period by evaluating the coefficients  $R_{yy}(m\Delta T)$  for  $m = 0, 1, 2, \dots, 10$ .

---

A sinusoidal signal of amplitude 1.4 mV and frequency 5 kHz is ‘buried’ in Gaussian noise with zero mean value. The noise has a uniform power spectral density of  $100 \text{ pW Hz}^{-1}$  up to a cut-off frequency of 1 MHz.

- (a) Find the total power, r.m.s. value and standard deviation for the noise signal.
- (b) What is the signal-to-noise ratio in dB?
- (c) Sketch the autocorrelation function for the combined signal and noise.
- (d) The combined signal is passed through a band-pass filter with centre frequency 5 kHz and bandwidth 1 kHz. What improvement in signal-to-noise ratio is obtained?
- (e) The filtered signal is then passed through a signal averager which averages corresponding samples of 100 sections of signal. What further improvement in signal-to-noise ratio is obtained?

$$100 \text{ pW/Hz} * 1 \text{ MHz} = 100 \text{ mW}$$

Signal power



A thermocouple giving a 10 mV d.c. output voltage is connected to a high impedance digital voltmeter some distance away. A difference in potential exists between earth at the thermocouple and earth at the voltmeter. Using the equivalent circuit given in Figure Prob. 4.

- (a) Calculate the r.m.s. values of the series mode and common mode interference voltages at the voltmeter input.
- (b) If the digital voltmeter has a common mode rejection ratio of 100 dB, find the minimum and maximum possible measured voltages.

